```
> restart;#Schwar;
> with(PDEtools):
> with(Physics):
> Setup(mathematicalnotation=true)
```

$$[mathematicalnotation = true] \tag{1}$$

```
> Setup(coordinatesystems=cartesian)
```

$$Systems\ of\ spacetime\ coordinates\ are:\quad \{X = (x, y, z, t)\}$$

$$[coordinatesystems = \{X\}] \tag{2}$$

```
> g_[ ];
```

$$g_{\mu,\,\nu} = \begin{bmatrix} -1 & 0 & 0 & 0 \\ 0 & -1 & 0 & 0 \\ 0 & 0 & -1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \tag{3}$$

```
> simplify(1/(1-2·G·m/(r·c^2))); simplify(1-2·G·m/(r·c^2)); simplify(1-2·r/(2·G·m/c^2));
  simplify((r c^2-G m) c^2 dr^2/(G m ((r c^2-G m) c^2/(G m)-2 G m)));
```

$$-\frac{r\,c^2}{-r\,c^2+2\,G\,m}$$

$$\frac{r\,c^2-2\,G\,m}{r\,c^2}$$

$$\frac{-r\,c^2+G\,m}{G\,m}$$

$$\frac{(-r\,c^2+G\,m)\,c^2\,dr^2}{-c^4\,r+2\,G^2\,m^2+G\,c^2\,m} \tag{4}$$

```
> ds2 := r^2 dtheta^2 + r^2 sin(θ)^2 dphi^2 - r·c^2·dr^2/(2 m G - r·c^2) - (r c^2 - 2 m G)/r · dt^2;
```

$$ds2 := r^2\,d\theta^2 + r^2\sin(\theta)^2\,d\phi^2 - \frac{r\,c^2\,dr^2}{-r\,c^2+2\,G\,m} - \frac{(r\,c^2-2\,G\,m)\,dt^2}{r} \tag{5}$$

```
> f(r,t) := r^2 dtheta^2 + r^2 sin(θ)^2 dphi^2 + r·c^2·dr^2/(r·c^2 - 2 m G) - (r c^2 - 2 m G)/r · dt^2;
```

$$f := (r,t) \mapsto r^2\,d\theta^2 + r^2\sin(\theta)^2\,d\phi^2 + r\,c^2\,dr^2\,\frac{1}{r\,c^2-2\,m\,G} - (r\,c^2-2\,m\,G)\,\frac{1}{r}\,dt^2 \tag{6}$$

```
> change_of_vars := {ρ = (m·G - r·c^2)/(m·G)}; simplify((-r c^2+G m) c^2 dr^2/(G m ((-r c^2+G m) c^2/(G m)-2 G m))); simplify(
  -((-r c^2+G m) c^2/(G m)-2 G m) G m dt^2/(-r c^2+G m)) #1-2 r/RScwar = 1-r/(mG/c^2) = (m·G-r·c^2)/(m·G)
```

$$change_of_vars := \left\{\rho = \frac{-r c^2 + G m}{G m}\right\}$$

$$-\frac{\left(-r c^2 + G m\right) c^2 dr^2}{c^4 r + 2 G^2 m^2 - G c^2 m}$$

$$\frac{\left(c^4 r + 2 G^2 m^2 - G c^2 m\right) dt^2}{-r c^2 + G m} \tag{7}$$

> $tmp := dchange(change_of_vars, [f(\rho, t)], [r, t], params = m, G, c);$

$$tmp := \left[\frac{\left(-r c^2 + G m\right)^2 d\theta^2}{G^2 m^2} + \frac{\left(-r c^2 + G m\right)^2 \sin(\theta)^2 d\phi^2}{G^2 m^2} + \frac{\left(-r c^2 + G m\right) c^2 dr^2}{G m \left(\frac{\left(-r c^2 + G m\right) c^2}{G m} - 2 G m\right)}\right.$$

$$\left. - \frac{\left(\frac{\left(-r c^2 + G m\right) c^2}{G m} - 2 G m\right) G m dt^2}{-r c^2 + G m}\right] \tag{8}$$

> $ds21 := \frac{\left(-r c^2 + m G\right)^2}{m^2 G^2} \cdot dtheta^2 + \frac{\left(-r \cdot c^2 + m G\right)^2}{G^2 m^2} \cdot \sin(\theta)^2 \cdot dphi^2 - \frac{\left(-r c^2 + G m\right) c^2}{c^4 r + 2 G^2 m^2 - G c^2 m} \cdot dr^2$
$+ \frac{\left(c^4 r - 2 G^2 m^2 - G c^2 m\right) \cdot dt^2}{-r c^2 + G m};$

$$ds21 := \frac{\left(-r c^2 + G m\right)^2 d\theta^2}{G^2 m^2} + \frac{\left(-r c^2 + G m\right)^2 \sin(\theta)^2 d\phi^2}{G^2 m^2} - \frac{\left(-r c^2 + G m\right) c^2 dr^2}{c^4 r + 2 G^2 m^2 - G c^2 m}$$

$$+ \frac{\left(c^4 r - 2 G^2 m^2 - G c^2 m\right) dt^2}{-r c^2 + G m} \tag{9}$$

> $Setup(coordinates = spherical, metric = ds2);$

Systems of spacetime coordinates are: $\{X = (r, \theta, \phi, t)\}$

Coordinates: $[r, \theta, \phi, t]$. *Signature:* $(- - - +)$

$$g_{\mu, \nu} = \begin{bmatrix} -\frac{r c^2}{-r c^2 + 2 G m} & 0 & 0 & 0 \\ 0 & r^2 & 0 & 0 \\ 0 & 0 & r^2 \sin(\theta)^2 & 0 \\ 0 & 0 & 0 & \frac{-r c^2 + 2 G m}{r} \end{bmatrix}$$

Setting lowercaselatin_is letters to represent space indices

$$\left[coordinatesystems = \{X\}, metric = \left\{(1, 1) = -\frac{r c^2}{-r c^2 + 2 G m}, (2, 2) = r^2, (3, 3) = r^2 \sin(\theta)^2, (4, 4) = -\frac{r c^2 - 2 G m}{r}\right\}, spaceindices = lowercaselatin_is\right] \tag{10}$$

> *Ricci*[*scalarsdefinition*];

$$\Phi_{00} = -\frac{R^{\mu,\nu}\, l_\mu l_\nu}{2},\ \Phi_{01} = -\frac{R^{\mu,\nu}\, l_\mu m_\nu}{2},\ \Phi_{02} = -\frac{R^{\mu,\nu}\, m_\mu m_\nu}{2},\ \Phi_{11} = -\frac{R^{\mu,\nu}\, \left(l_\mu n_\nu + m_\mu \overline{m}_\nu\right)}{4},\ \Phi_{12} = -\frac{R^{\mu,\nu}\, m_\mu n_\nu}{2},\ \Phi_{22} = -\frac{R^{\mu,\nu}\, n_\mu n_\nu}{2},\ \Lambda = \frac{R_\mu{}^\mu}{24} \tag{11}$$

> *Ricci*[*scalars*];

$$\Phi_{00} = 0,\ \Phi_{01} = 0,\ \Phi_{02} = 0,\ \Phi_{11} = 0,\ \Phi_{12} = 0,\ \Phi_{22} = 0,\ \Lambda = 0 \tag{12}$$

> *Ricci*[*scalar*];

$$0 \tag{13}$$

> *Ricci*[~μ, ν, *matrix*];

$$R^\mu{}_\nu = \begin{bmatrix} 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix} \tag{14}$$

> $\left(\frac{1}{2}\cdot g_{[\,]}\cdot Ricci[scalar] - Ricci[\sim\mu, \nu, matrix] + \Lambda\cdot g_{[\,]}\right)$;

for Λ=0,Ricci scalar&tensor = 0 the result is null

$$\Lambda\, g_{\mu,\nu} - R^\mu{}_\nu = \begin{bmatrix} -\frac{\Lambda\, r\, c^2}{-r\, c^2 + 2\, G\, m} & 0 & 0 & 0 \\ 0 & \Lambda\, r^2 & 0 & 0 \\ 0 & 0 & \Lambda\, r^2 \sin(\theta)^2 & 0 \\ 0 & 0 & 0 & \frac{\Lambda\left(-r\, c^2 + 2\, G\, m\right)}{r} \end{bmatrix} \tag{15}$$

> *EnergyMomentum*[ ]

$$T_{\mu,\nu} = \begin{bmatrix} 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix} \tag{16}$$

> *Setup*(*coordinates* = *spherical*, *metric* = *ds21*);

Systems of spacetime coordinates are: $\{X = (r, \theta, \phi, t)\}$

Coordinates: $[r, \theta, \phi, t]$. *Signature:* $(- - - +)$

$$g_{\mu,\nu} = \left[\left[-\frac{\left(-r\, c^2 + G\, m\right) c^2}{r\, c^4 + 2\, G^2\, m^2 - c^2\, G\, m}, 0, 0, 0\right], \left[0, \frac{\left(-r\, c^2 + G\, m\right)^2}{G^2\, m^2}, 0, 0\right], \left[0, 0, \frac{\left(-r\, c^2 + G\, m\right)^2 \sin(\theta)^2}{G^2\, m^2}, 0\right],$$

$$\left[0, 0, 0, \frac{r c^4 - 2 G^2 m^2 - c^2 G m}{-r c^2 + G m}\right]\Bigg]$$

$$\left[coordinatesystems = \{X\}, metric = \left\{(1,1) = -\frac{\left(-r c^2 + G m\right) c^2}{r c^4 + 2 G^2 m^2 - c^2 G m}, (2,2) = \frac{\left(-r c^2 + G m\right)^2}{G^2 m^2}, (3,3) \right. \right. \quad (17)$$

$$\left. \left. = \frac{\left(-r c^2 + G m\right)^2 \sin(\theta)^2}{G^2 m^2}, (4,4) = \frac{r c^4 - 2 G^2 m^2 - c^2 G m}{-r c^2 + G m}\right\}\right]$$

> *Ricci*[*scalarsdefinition*];

$$\Phi_{00} = -\frac{R^{\mu,\nu}\, l_\mu\, l_\nu}{2}, \Phi_{01} = -\frac{R^{\mu,\nu}\, l_\mu\, m_\nu}{2}, \Phi_{02} = -\frac{R^{\mu,\nu}\, m_\mu\, m_\nu}{2}, \Phi_{11} = -\frac{R^{\mu,\nu}\left(l_\mu\, n_\nu + m_\mu\, \overline{m}_\nu\right)}{4}, \Phi_{12} = \quad (18)$$

$$-\frac{R^{\mu,\nu}\, m_\mu\, n_\nu}{2}, \Phi_{22} = -\frac{R^{\mu,\nu}\, n_\mu\, n_\nu}{2}, \Lambda = \frac{R_\mu{}^\mu}{24}$$

> *Ricci*[*scalars*];

$$\Phi_{00} = -\frac{c^4 G^2 m^2}{\left(-r c^2 + G m\right)^2 \left(-r c^4 + 2 G^2 m^2 + c^2 G m\right)}, \Phi_{01} = 0, \Phi_{02} = 0, \Phi_{11} \quad (19)$$

$$= \frac{1}{16\left(-r c^2 + G m\right)^2 \left(-\frac{1}{2} r c^4 + G^2 m^2 + \frac{1}{2} c^2 G m\right)^2}\left(4 G^6 m^6 + 4 G^5 c^2 m^5 - 4\left(r - \frac{3}{4}\right) m^4 c^4 G^4\right.$$

$$\left. - 2 G^3 c^6 m^3 r + c^8 m^2 (r-1)(r+1) G^2 + 2 G c^{10} m r - c^{12} r^2\right), \Phi_{12} = 0, \Phi_{22} =$$

$$-\frac{c^4 G^2 m^2}{\left(-r c^2 + G m\right)^2 \left(-r c^4 + 2 G^2 m^2 + c^2 G m\right)}, \Lambda$$

$$= \frac{1}{48\left(-r c^2 + G m\right)^2 \left(-\frac{1}{2} r c^4 + G^2 m^2 + \frac{1}{2} c^2 G m\right)^2}\left(4 G^6 m^6 + 4 G^5 c^2 m^5 - 4 m^4 c^4\left(r + \frac{1}{4}\right) G^4\right.$$

$$\left. - 2 c^6 m^3 (r+2) G^3 + c^8 m^2 \left(r^2 + 4 r - 1\right) G^2 + 2 G c^{10} m r - c^{12} r^2\right)$$

> *Ricci*[*scalar*];

$$\frac{1}{\left(-r c^2 + G m\right)^2 \left(-r c^4 + 2 G^2 m^2 + c^2 G m\right)^2}\left(2\left(G^2 c^8 m^2 r^2 - c^{12} r^2 - 4 G^4 c^4 m^4 r - 2 G^3 c^6 m^3 r\right.\right. \quad (20)$$

$$\left.\left. + 4 G^2 c^8 m^2 r + 2 G c^{10} m r + 4 G^6 m^6 + 4 G^5 c^2 m^5 - c^4 G^4 m^4 - 4 G^3 c^6 m^3 - G^2 c^8 m^2\right)\right)$$

> *with*(*LinearAlgebra*) :

> *simplify*(*g_*[1, 1]); *simplify*(*g_*[4, 4]);

$$-\frac{\left(-r c^2 + G m\right) c^2}{r c^4 + 2 G^2 m^2 - c^2 G m}$$

$$\frac{r c^4 - 2 G^2 m^2 - c^2 G m}{-r c^2 + G m} \quad (21)$$

> $R_\mu{}^\mu$; *g_*[ ];

$$R_\mu{}^\mu$$

$$g_{\mu,\nu} \quad (22)$$

$$= \left[\left[-\frac{\left(-r c^2 + G m\right) c^2}{r c^4 + 2 G^2 m^2 - c^2 G m}, 0, 0, 0 \right], \left[0, \frac{\left(-r c^2 + G m\right)^2}{G^2 m^2}, 0, 0\right], \left[0, 0, \frac{\left(-r c^2 + G m\right)^2 \sin(\theta)^2}{G^2 m^2}, 0\right], \left[0, 0, 0, \frac{r c^4 - 2 G^2 m^2 - c^2 G m}{-r c^2 + G m}\right]\right]$$

> $TR := r = \dfrac{m \cdot G - \rho \cdot c^2}{m G}$;

$$TR := r = \frac{-\rho c^2 + G m}{G m} \tag{23}$$

> $subs(r = \mathrm{rho}, TransformCoordinates(TR, g_[\mathrm{mu}, \mathrm{nu}], [\mathrm{rho}, \mathrm{theta}, \mathrm{phi}, t]))$;

$$\left[\left[-\frac{c^6\left(c^4 \rho + G^2 m^2 - c^2 G m\right)}{G^2 m^2\left(-c^6 \rho + 2 G^3 m^3 - c^2 G^2 m^2 + G c^4 m\right)}, 0, 0, 0 \right], \left[0, \frac{\left(c^4 \rho + G^2 m^2 - c^2 G m\right)^2}{G^4 m^4}, 0, 0\right], \left[0, 0, \frac{\left(c^4 \rho + G^2 m^2 - c^2 G m\right)^2 \sin(\theta)^2}{G^4 m^4}, 0\right], \left[0, 0, 0, \frac{-c^6 \rho - 2 G^3 m^3 - c^2 G^2 m^2 + G c^4 m}{c^4 \rho + G^2 m^2 - c^2 G m}\right]\right] \tag{24}$$

> $subs(\rho = r, TransformCoordinates(TR, g_[\mathrm{mu}, \mathrm{nu}], [\mathrm{rho}, \mathrm{theta}, \mathrm{phi}, t], output = line_element))$;

$$-\frac{1}{G^4 m^4\left(-c^6 r + 2 G^3 m^3 - c^2 G^2 m^2 + G c^4 m\right)}\left(\left(G^2 (\mathrm{d}(r))^2 c^6 m^2 - \left(-c^6 r + 2 G^3 m^3 - c^2 G^2 m^2 + G c^4 m\right)\left(r c^4 + G^2 m^2 - c^2 G m\right)\left(\sin(\theta)^2 (\mathrm{d}(\phi))^2 + (\mathrm{d}(\theta))^2\right)\right)\left(r c^4 + G^2 m^2 - c^2 G m\right)\right) + \frac{\left(-c^6 r - 2 G^3 m^3 - c^2 G^2 m^2 + G c^4 m\right)(\mathrm{d}(t))^2}{r c^4 + G^2 m^2 - c^2 G m} \tag{25}$$

> $ds2 := \dfrac{\left(r c^4 + G^2 m^2 - c^2 G m\right)^2}{G^4 m^4} dtheta^2 + \dfrac{\left(r c^4 + G^2 m^2 - c^2 G m\right)^2}{G^4 m^4} \sin(\theta)^2 dphi^2 + \dfrac{G^2 \cdot c^6 m^2 \cdot \left(r c^4 + G^2 m^2 - c^2 G m\right)}{G^4 m^4\left(-c^6 r + 2 G^3 m^3 - c^2 G^2 m^2 + G c^4 m\right)} dr^2 + \dfrac{\left(-c^6 r - 2 G^3 m^3 - c^2 G^2 m^2 + G c^4 m\right)}{r c^4 + G^2 m^2 - c^2 G m} \cdot dt^2$;

$$ds2 := \frac{\left(r c^4 + G^2 m^2 - c^2 G m\right)^2 d\theta^2}{G^4 m^4} + \frac{\left(r c^4 + G^2 m^2 - c^2 G m\right)^2 \sin(\theta)^2 d\phi^2}{G^4 m^4} + \frac{c^6\left(r c^4 + G^2 m^2 - c^2 G m\right) dr^2}{G^2 m^2\left(-c^6 r + 2 G^3 m^3 - c^2 G^2 m^2 + G c^4 m\right)} + \frac{\left(-c^6 r - 2 G^3 m^3 - c^2 G^2 m^2 + G c^4 m\right) dt^2}{r c^4 + G^2 m^2 - c^2 G m} \tag{26}$$

>

> $Setup(coordinates = spherical, metric = ds2)$;

Systems of spacetime coordinates are: $\{X = (r, \theta, \phi, t)\}$

Coordinates: $[r, \theta, \phi, t]$. *Signature:* $(- - - +)$

$$g_{\mu,\nu} = \left[\left[\frac{c^6\left(r c^4 + G^2 m^2 - c^2 G m\right)}{-G^2 c^6 m^2 r + 2 G^5 m^5 - G^4 c^2 m^4 + G^3 c^4 m^3}, 0, 0, 0\right], \left[0, \frac{\left(r c^4 + G^2 m^2 - c^2 G m\right)^2}{G^4 m^4}, 0, 0\right], \left[0, 0, \frac{\left(r c^4 + G^2 m^2 - c^2 G m\right)^2 \sin(\theta)^2}{G^4 m^4}, 0\right], \left[0, 0, 0, \frac{-c^6 r - 2 G^3 m^3 - c^2 G^2 m^2 + G c^4 m}{r c^4 + G^2 m^2 - c^2 G m}\right]\right]$$

$$\left[coordinatesystems = \{X\}, metric = \left\{(1,1) = \frac{c^6\left(r c^4 + G^2 m^2 - c^2 G m\right)}{-G^2 c^6 m^2 r + 2 G^5 m^5 - G^4 c^2 m^4 + G^3 c^4 m^3}, (2,2) = \frac{\left(r c^4 + G^2 m^2 - c^2 G m\right)^2}{G^4 m^4}, (3,3) = \frac{\left(r c^4 + G^2 m^2 - c^2 G m\right)^2 \sin(\theta)^2}{G^4 m^4}, (4,4) = \frac{-c^6 r - 2 G^3 m^3 - c^2 G^2 m^2 + G c^4 m}{r c^4 + G^2 m^2 - c^2 G m}\right\}\right] \quad (27)$$

> Ricci[scalarsdefinition];

$$\Phi_{00} = -\frac{R^{\mu,\nu} l_\mu l_\nu}{2}, \Phi_{01} = -\frac{R^{\mu,\nu} l_\mu m_\nu}{2}, \Phi_{02} = -\frac{R^{\mu,\nu} m_\mu m_\nu}{2}, \Phi_{11} = -\frac{R^{\mu,\nu}\left(l_\mu n_\nu + m_\mu \overline{m}_\nu\right)}{4}, \Phi_{12} = -\frac{R^{\mu,\nu} m_\mu n_\nu}{2}, \Phi_{22} = -\frac{R^{\mu,\nu} n_\mu n_\nu}{2}, \Lambda = \frac{R^\mu{}_\mu}{24} \quad (28)$$

> Ricci[scalars];

$$\Phi_{00} = \frac{G^5 m^5 c^4}{\left(c^6 r + 2 G^3 m^3 + c^2 G^2 m^2 - G c^4 m\right)\left(r c^4 + G^2 m^2 - c^2 G m\right)^2}, \Phi_{01} = 0, \Phi_{02} = 0, \Phi_{11} = \left(m^2 G^2\left(m^8 G^8 + G^7 c^2 m^7 - \frac{5 G^6 m^6 c^4}{4} + c^6 m^5\left(r - \frac{1}{2}\right) G^5 + \frac{c^8 m^4 (r+1) G^4}{2} - \frac{c^{10} m^3 (r+1) G^3}{2} + \frac{c^{12} m^2 (r+1)^2 G^2}{4} - \frac{G c^{14} m r}{2} + \frac{c^{16} r^2}{4}\right)\right) \Big/ \left(4\left(\frac{1}{2} c^6 r + G^3 m^3 + \frac{1}{2} c^2 G^2 m^2 - \frac{1}{2} G c^4 m\right)^2 \left(r c^4 + G^2 m^2 - c^2 G m\right)^2\right), \Phi_{12} = 0, \Phi_{22} = \frac{G^5 m^5 c^4}{\left(c^6 r + 2 G^3 m^3 + c^2 G^2 m^2 - G c^4 m\right)\left(r c^4 + G^2 m^2 - c^2 G m\right)^2}, \Lambda = \left(m^2\left(m^8 G^8 + G^7 c^2 m^7 - \frac{G^6 m^6 c^4}{4} + c^6 m^5\left(r + \frac{1}{2}\right) G^5 + \frac{c^8 m^4 (r-1) G^4}{2} + \frac{c^{10} m^3 (r-1) G^3}{2} + \frac{c^{12} m^2 (r+1)^2 G^2}{4} - \frac{G c^{14} m r}{2} + \frac{c^{16} r^2}{4}\right) G^2\right) \Big/ \left(12\left(\frac{1}{2} c^6 r + G^3 m^3 + \frac{1}{2} c^2 G^2 m^2 - \frac{1}{2} G c^4 m\right)^2 \left(r c^4 + G^2 m^2 - c^2 G m\right)^2\right) \quad (29)$$

> Ricci[scalar];

$$\left(2\,G^2\,m^2\left(G^2\,c^{12}\,m^2\,r^2+c^{16}\,r^2+4\,G^5\,c^6\,m^5\,r+2\,G^4\,c^8\,m^4\,r+2\,G^3\,c^{10}\,m^3\,r+2\,G^2\,c^{12}\,m^2\,r-2\,G\,c^{14}\,m\,r\right.\right. \tag{30}$$
$$\left.\left.+4\,m^8\,G^8+4\,G^7\,c^2\,m^7-G^6\,m^6\,c^4+2\,G^5\,c^6\,m^5-2\,G^4\,c^8\,m^4-2\,G^3\,c^{10}\,m^3+G^2\,c^{12}\,m^2\right)\right)\Big/\left(\left(r\,c^4\right.\right.$$
$$\left.\left.+G^2\,m^2-c^2\,G\,m\right)^2\left(c^6\,r+2\,G^3\,m^3+c^2\,G^2\,m^2-G\,c^4\,m\right)^2\right)$$

> g_[ ];

$$g_{\mu,\,\nu}=\left[\left[\frac{c^6\left(r\,c^4+G^2\,m^2-c^2\,G\,m\right)}{-G^2\,c^6\,m^2\,r+2\,G^5\,m^5-G^4\,c^2\,m^4+G^3\,c^4\,m^3},\,0,\,0,\,0\right],\right. \tag{31}$$
$$\left[0,\,\frac{\left(r\,c^4+G^2\,m^2-c^2\,G\,m\right)^2}{G^4\,m^4},\,0,\,0\right],$$
$$\left[0,\,0,\,\frac{\left(r\,c^4+G^2\,m^2-c^2\,G\,m\right)^2\sin(\theta)^2}{G^4\,m^4},\,0\right],$$
$$\left.\left[0,\,0,\,0,\,\frac{-c^6\,r-2\,G^3\,m^3-c^2\,G^2\,m^2+G\,c^4\,m}{r\,c^4+G^2\,m^2-c^2\,G\,m}\right]\right]$$

> $\left(\frac{1}{2}\cdot g_{[\,]}\cdot Ricci[scalar]-Ricci[\sim\mu,\nu,matrix]+\Lambda\cdot g_{[\,]}\right)$;

$$\left(g_{\mu,\,\nu}\,G^2\,m^2\left(G^2\,c^{12}\,m^2\,r^2+c^{16}\,r^2+4\,G^5\,c^6\,m^5\,r+2\,G^4\,c^8\,m^4\,r+2\,G^3\,c^{10}\,m^3\,r+2\,G^2\,c^{12}\,m^2\,r-2\,G\,c^{14}\,m\,r\right.\right. \tag{32}$$
$$\left.\left.+4\,m^8\,G^8+4\,G^7\,c^2\,m^7-G^6\,m^6\,c^4+2\,G^5\,c^6\,m^5-2\,G^4\,c^8\,m^4-2\,G^3\,c^{10}\,m^3+G^2\,c^{12}\,m^2\right)\right)\Big/\left(\left(r\,c^4\right.\right.$$
$$\left.\left.+G^2\,m^2-c^2\,G\,m\right)^2\left(c^6\,r+2\,G^3\,m^3+c^2\,G^2\,m^2-G\,c^4\,m\right)^2\right)-R^{\mu}{}_{\nu}+\Lambda\,g_{\mu,\,\nu}=\left[\right.$$
$$\left[\left(G^2\,m^2\left(G^2\,c^{12}\,m^2\,r^2+c^{16}\,r^2+4\,G^5\,c^6\,m^5\,r+2\,G^4\,c^8\,m^4\,r+2\,G^3\,c^{10}\,m^3\,r+2\,G^2\,c^{12}\,m^2\,r-2\,G\,c^{14}\,m\,r\right.\right.\right.$$
$$\left.\left.\left.+4\,m^8\,G^8+4\,G^7\,c^2\,m^7-G^6\,m^6\,c^4+2\,G^5\,c^6\,m^5-2\,G^4\,c^8\,m^4-2\,G^3\,c^{10}\,m^3+G^2\,c^{12}\,m^2\right)\,c^6\right)\Big/\left(\left(r\,c^4\right.\right.\right.$$
$$\left.\left.\left.+G^2\,m^2-c^2\,G\,m\right)\left(c^6\,r+2\,G^3\,m^3+c^2\,G^2\,m^2-G\,c^4\,m\right)^2\left(-G^2\,c^6\,m^2\,r+2\,G^5\,m^5-G^4\,c^2\,m^4\right.\right.\right.$$
$$\left.\left.\left.+G^3\,c^4\,m^3\right)\right)-\frac{4\,G^5\,c^{10}\,m^5\,r+6\,G^8\,m^8\,c^4+4\,G^7\,c^6\,m^7-4\,c^8\,m^6\,G^6}{\left(r\,c^4+G^2\,m^2-c^2\,G\,m\right)^2\left(c^6\,r+2\,G^3\,m^3+c^2\,G^2\,m^2-G\,c^4\,m\right)^2}$$
$$\left.+\frac{\Lambda\,c^6\left(r\,c^4+G^2\,m^2-c^2\,G\,m\right)}{-G^2\,c^6\,m^2\,r+2\,G^5\,m^5-G^4\,c^2\,m^4+G^3\,c^4\,m^3},\,0,\,0,\,0\right],$$
$$\left[0,\,\frac{1}{G^2\,m^2\left(c^6\,r+2\,G^3\,m^3+c^2\,G^2\,m^2-G\,c^4\,m\right)^2}\left(G^2\,c^{12}\,m^2\,r^2+c^{16}\,r^2+4\,G^5\,c^6\,m^5\,r\right.\right.$$
$$\left.+2\,G^4\,c^8\,m^4\,r+2\,G^3\,c^{10}\,m^3\,r+2\,G^2\,c^{12}\,m^2\,r-2\,G\,c^{14}\,m\,r+4\,m^8\,G^8+4\,G^7\,c^2\,m^7-G^6\,m^6\,c^4\right.$$
$$\left.\left.+2\,G^5\,c^6\,m^5-2\,G^4\,c^8\,m^4-2\,G^3\,c^{10}\,m^3+G^2\,c^{12}\,m^2\right)\right.$$

$$-\frac{2\left(G^5 m^5+\frac{G^4 c^2 m^4}{2}-\frac{G^3 c^4 m^3}{2}+\frac{c^6 m^2 (r+1) G^2}{2}-\frac{G c^8 m}{2}+\frac{c^{10} r}{2}\right) m^2 G^2}{\left(c^6 r+2 G^3 m^3+c^2 G^2 m^2-G c^4 m\right)\left(r c^4+G^2 m^2-c^2 G m\right)^2}$$

$$+\frac{\Lambda\left(r c^4+G^2 m^2-c^2 G m\right)^2}{G^4 m^4}, 0, 0\Bigg],$$

$$\Bigg[0, 0, \frac{1}{G^2 m^2\left(c^6 r+2 G^3 m^3+c^2 G^2 m^2-G c^4 m\right)^2}\Big(\left(G^2 c^{12} m^2 r^2+c^{16} r^2+4 G^5 c^6 m^5 r\right.$$

$$\left.+2 G^4 c^8 m^4 r+2 G^3 c^{10} m^3 r+2 G^2 c^{12} m^2 r-2 G c^{14} m r+4 m^8 G^8+4 G^7 c^2 m^7-G^6 m^6 c^4\right.$$

$$\left.+2 G^5 c^6 m^5-2 G^4 c^8 m^4-2 G^3 c^{10} m^3+G^2 c^{12} m^2\right)\sin(\theta)^2\Big)$$

$$-\frac{2\left(G^5 m^5+\frac{G^4 c^2 m^4}{2}-\frac{G^3 c^4 m^3}{2}+\frac{c^6 m^2 (r+1) G^2}{2}-\frac{G c^8 m}{2}+\frac{c^{10} r}{2}\right) m^2 G^2}{\left(c^6 r+2 G^3 m^3+c^2 G^2 m^2-G c^4 m\right)\left(r c^4+G^2 m^2-c^2 G m\right)^2}$$

$$+\frac{\Lambda\left(r c^4+G^2 m^2-c^2 G m\right)^2\sin(\theta)^2}{G^4 m^4}, 0\Bigg],$$

$$\Bigg[0, 0, 0, \Big(G^2 m^2\Big(G^2 c^{12} m^2 r^2+c^{16} r^2+4 G^5 c^6 m^5 r+2 G^4 c^8 m^4 r+2 G^3 c^{10} m^3 r+2 G^2 c^{12} m^2 r$$

$$-2 G c^{14} m r+4 m^8 G^8+4 G^7 c^2 m^7-G^6 m^6 c^4+2 G^5 c^6 m^5-2 G^4 c^8 m^4-2 G^3 c^{10} m^3+G^2 c^{12} m^2\Big)$$

$$\left(-c^6 r-2 G^3 m^3-c^2 G^2 m^2+G c^4 m\right)\Big)\Big/\Big(\left(r c^4+G^2 m^2-c^2 G m\right)^3\left(c^6 r+2 G^3 m^3+c^2 G^2 m^2\right.$$

$$\left.-G c^4 m\right)^2\Big)+\frac{2 G^8 m^8 c^4}{\left(r c^4+G^2 m^2-c^2 G m\right)^2\left(c^6 r+2 G^3 m^3+c^2 G^2 m^2-G c^4 m\right)^2}$$

$$+\frac{\Lambda\left(-c^6 r-2 G^3 m^3-c^2 G^2 m^2+G c^4 m\right)}{r c^4+G^2 m^2-c^2 G m}\Bigg]\Bigg]$$

```
> restart;#
> with(Physics):
> Setup(mathematicalnotation = true)
```

$$[mathematicalnotation = true] \quad (33)$$

```
> Setup(coordinatesystems = cartesian)
```

$$Systems\ of\ spacetime\ coordinates\ are:\ \{X = (x, y, z, t)\}$$

$$[coordinatesystems = \{X\}] \quad (34)$$

$$[mathematicalnotation = true] \quad (35)$$

```
> ds2 := r^2 dtheta^2 + r^2 sin(theta)^2 dphi^2 - r·c^2·dr^2/(2 m G - r·c^2) - (r c^2 - 2 m G)/r · dt^2;
```

$$ds2 := r^2\,d\theta^2 + r^2\sin(\theta)^2\,d\phi^2 - \frac{r\,c^2\,dr^2}{-r\,c^2 + 2\,m\,G} - \frac{(r\,c^2 - 2\,m\,G)\,dt^2}{r} \tag{36}$$

> `Setup(coordinates = spherical, metric = ds2);`

Systems of spacetime coordinates are: $\{X = (r, \theta, \phi, t)\}$

Coordinates: $[r, \theta, \phi, t]$. *Signature:* $(- - - +)$

$$g_{\mu,\nu} = \begin{bmatrix} -\frac{r\,c^2}{-r\,c^2 + 2\,m\,G} & 0 & 0 & 0 \\ 0 & r^2 & 0 & 0 \\ 0 & 0 & r^2\sin(\theta)^2 & 0 \\ 0 & 0 & 0 & \frac{-r\,c^2 + 2\,m\,G}{r} \end{bmatrix}$$

Setting lowercaselatin_is letters to represent space indices

$$\left[coordinatesystems = \{X\},\ metric = \left\{(1,1) = -\frac{r\,c^2}{-r\,c^2 + 2\,m\,G},\ (2,2) = r^2,\ (3,3) = r^2\sin(\theta)^2,\ (4,4) = -\frac{r\,c^2 - 2\,m\,G}{r}\right\},\ spaceindices = lowercaselatin_is\right] \tag{37}$$

> $TR := r = \frac{m\cdot G - \rho\cdot c^2}{m\,G}$;

$$TR := r = \frac{-\rho\,c^2 + m\,G}{m\,G} \tag{38}$$

> `subs(r = rho, TransformCoordinates(TR, g_[mu, nu], [rho, theta, phi, t]));`

$$\left[\left[-\frac{c^6\,(-\rho\,c^2 + m\,G)}{m^2\,G^2\,(c^4\,\rho + 2\,m^2\,G^2 - c^2\,m\,G)}, 0, 0, 0\right], \left[0, \frac{(-\rho\,c^2 + m\,G)^2}{m^2\,G^2}, 0, 0\right], \left[0, 0, \frac{(-\rho\,c^2 + m\,G)^2\sin(\theta)^2}{m^2\,G^2}, 0\right], \left[0, 0, 0, \frac{c^4\,\rho + 2\,m^2\,G^2 - c^2\,m\,G}{-\rho\,c^2 + m\,G}\right]\right] \tag{39}$$

> `subs(ρ = r, TransformCoordinates(TR, g_[mu, nu], [rho, theta, phi, t], output = line_element));`

$$-\frac{1}{m^2\,G^2\,(c^4\,r + 2\,m^2\,G^2 - G\,c^2\,m)}\left((-r\,c^2 + m\,G)\left((\mathrm{d}(r))^2\,c^6 - (c^4\,r + 2\,m^2\,G^2 - G\,c^2\,m)\,(-r\,c^2 + m\,G)\left(\sin(\theta)^2\,(\mathrm{d}(\phi))^2 + (\mathrm{d}(\theta))^2\right)\right)\right) + \frac{(c^4\,r + 2\,m^2\,G^2 - G\,c^2\,m)\,(\mathrm{d}(t))^2}{-r\,c^2 + m\,G} \tag{40}$$

> $ds22 := \frac{(-r\,c^2 + m\,G)^2}{m^2\,G^2}\,dtheta^2 + \frac{(-r\,c^2 + m\,G)^2}{m^2\,G^2}\sin(\theta)^2\,dphi^2 - \frac{c^6\,(-r\,c^2 + m\,G)}{m^2\,G^2\,(c^4\,r + 2\,m^2\,G^2 - c^2\,m\,G)}$

$$\cdot dr^2 + \frac{\left(c^4 r + 2 m^2 G^2 - G c^2 m\right)}{-r c^2 + m G} \cdot dt^2 ;$$

$$ds22 := \frac{\left(-r c^2 + m G\right)^2 d\theta^2}{m^2 G^2} + \frac{\left(-r c^2 + m G\right)^2 \sin(\theta)^2 d\phi^2}{m^2 G^2} - \frac{c^6 \left(-r c^2 + m G\right) dr^2}{m^2 G^2 \left(c^4 r + 2 m^2 G^2 - G c^2 m\right)} + \frac{\left(c^4 r + 2 m^2 G^2 - G c^2 m\right) dt^2}{-r c^2 + m G} \quad (41)$$

> $Setup(coordinates = spherical, metric = ds22);$

Systems of spacetime coordinates are: $\{X = (r, \theta, \phi, t)\}$

Coordinates: $[r, \theta, \phi, t]$. *Signature:* $(- - - +)$

$$g_{\mu,\nu} = \left[\left[-\frac{c^6 \left(-r c^2 + m G\right)}{m^2 G^2 \left(c^4 r + 2 m^2 G^2 - G c^2 m\right)}, 0, 0, 0\right], \left[0, \frac{\left(-r c^2 + m G\right)^2}{m^2 G^2}, 0, 0\right], \left[0, 0, \frac{\left(-r c^2 + m G\right)^2 \sin(\theta)^2}{m^2 G^2}, 0\right], \left[0, 0, 0, \frac{c^4 r + 2 m^2 G^2 - G c^2 m}{-r c^2 + m G}\right]\right]$$

$$\left[coordinatesystems = \{X\}, metric = \left\{(1,1) = -\frac{c^6 \left(-r c^2 + m G\right)}{m^2 G^2 \left(c^4 r + 2 m^2 G^2 - G c^2 m\right)}, (2,2) = \frac{\left(-r c^2 + m G\right)^2}{m^2 G^2}, (3,3) = \frac{\left(-r c^2 + m G\right)^2 \sin(\theta)^2}{m^2 G^2}, (4,4) = \frac{c^4 r + 2 m^2 G^2 - G c^2 m}{-r c^2 + m G}\right\}\right] \quad (42)$$

> $Ricci[scalarsdefinition];$

$$\Phi_{00} = -\frac{R^{\mu,\nu} l_\mu l_\nu}{2}, \Phi_{01} = -\frac{R^{\mu,\nu} l_\mu m_\nu}{2}, \Phi_{02} = -\frac{R^{\mu,\nu} m_\mu m_\nu}{2}, \Phi_{11} = -\frac{R^{\mu,\nu} \left(l_\mu n_\nu + m_\mu \overline{m}_\nu\right)}{4}, \Phi_{12} = -\frac{R^{\mu,\nu} m_\mu n_\nu}{2}, \Phi_{22} = -\frac{R^{\mu,\nu} n_\mu n_\nu}{2}, \Lambda = \frac{R^\mu}{24} \quad (43)$$

> $Ricci[scalars];$

$$\Phi_{00} = 0, \Phi_{01} = 0, \Phi_{02} = 0, \Phi_{11} = 0, \Phi_{12} = 0, \Phi_{22} = 0, \Lambda = 0 \quad (44)$$

> $Ricci[scalar]$

$$0 \quad (45)$$

> $Ricci[\sim\mu, \nu, matrix]$

$$R^\mu{}_\nu = \begin{bmatrix} 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix} \quad (46)$$

> $g_[\ ];$

$$g_{\mu,\nu} = \left[\left[-\frac{c^6\left(-r c^2 + m G\right)}{m^2 G^2\left(c^4 r + 2 m^2 G^2 - G c^2 m\right)}, 0, 0, 0\right], \left[0, \frac{\left(-r c^2 + m G\right)^2}{m^2 G^2}, 0, 0\right], \left[0, 0, \frac{\left(-r c^2 + m G\right)^2 \sin(\theta)^2}{m^2 G^2}, 0\right], \left[0, 0, 0, \frac{c^4 r + 2 m^2 G^2 - G c^2 m}{-r c^2 + m G}\right]\right] \quad (47)$$

> $\left(\frac{1}{2}\cdot g_{[\,]}\cdot Ricci[scalar] - Ricci[\sim\mu, \nu, matrix] + \Lambda\cdot g_{[\,]}\right)$;

for Λ=0 Ricci scalar and tensor=0 the result is null

$$\Lambda\, g_{\mu,\nu} - R^{\mu}{}_{\nu} = \left[\left[-\frac{\Lambda\, c^6\left(-r c^2 + m G\right)}{m^2 G^2\left(c^4 r + 2 m^2 G^2 - G c^2 m\right)}, 0, 0, 0\right], \left[0, \frac{\Lambda\left(-r c^2 + m G\right)^2}{m^2 G^2}, 0, 0\right], \left[0, 0, \frac{\Lambda\left(-r c^2 + m G\right)^2 \sin(\theta)^2}{m^2 G^2}, 0\right], \left[0, 0, 0, \frac{\Lambda\left(c^4 r + 2 m^2 G^2 - G c^2 m\right)}{-r c^2 + m G}\right]\right] \quad (48)$$

$$(49)$$

> *EnergyMomentum*[*definition*];

$$T_{\mu,\nu} = \frac{G_{\mu,\nu}}{8\pi} \quad (50)$$

>